

Solution Requirements

Date	July7, 2026
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Project Name	Caption AI: AI-Powered Social Media Caption Generator

Functional and Non-Functional Requirements:

The following requirements define the features and quality standards of the CaptionAI application.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Access	Users can access the application directly through the web interface without creating an account or logging in.	Low
2	Platform Selection	Users can select the desired social media platform such as Instagram, LinkedIn, or Twitter (X) before generating content.	High
3	AI Content Generation	The system sends the user's request to the Groq API and generates AI-powered captions, hashtags, and CTA suggestions.	High
4	Request Processing	The Flask backend receives the user input, validates it, and processes the request before sending it to the AI model.	High
5	Results Display	The generated captions, hashtags, and CTA suggestions are displayed clearly on the user interface.	High
6	Input Validation	The system checks that the user has entered a campaign description and selected the required options before processing the request.	High
7	Security	The Groq API key is stored securely using the .env file and is never exposed in the frontend code.	High
8	Copy Functionality	Users can copy the generated captions, hashtags, and CTA suggestions with a single click.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance	The application should generate AI-powered content quickly after receiving the user's request.	Content should normally be displayed within a few seconds, depending on network speed and API response time.
2	Scalability	The application should support future enhancements such as additional social media platforms and new AI features.	The project should be easy to maintain and extend.
3	Security	API credentials must be stored securely and protected from public access.	API keys remain in the .env file and are not uploaded to GitHub.
4	Reliability	The application should provide consistent results and handle requests without unexpected failures.	The application should respond successfully under normal operating conditions.
5	Usability	The interface should be simple, responsive, and easy to use for both desktop and mobile users.	Users should be able to generate and copy content without difficulty.
6	Compatibility	The application should run on standard web browsers without requiring special hardware or software.	Compatible with modern browsers and Python 3.x environment.